Reasoning test

For admission to Class 9

About the test

- 30 questions in 30 minutes: 10 each of maths, English and science, mixed together.
- Half the maths and science questions are from the Class 8 syllabus (the chapters listed below); the rest test reasoning. Each question is marked Basics or Class 8.
- Taken online on a phone or computer, through a link sent to the parent's WhatsApp. The link works for 2 hours.
- Most questions have four options; a few ask for a number to be typed in.
- One mark for a correct answer. A quarter mark off for a wrong multiple-choice answer; nothing off for a wrong type-in answer or a blank.
- Answers can be changed before submitting. If the candidate leaves the page, the timer pauses.

What the test covers: basics

Basics questions test reasoning, not memory, using ideas taught up to Class 8. Chapters are from the NCERT (CBSE) books a student joining Class 9 in 2027 will have used; other boards teach the same ideas under their own names.

Maths

Class 6 · Ganita Prakash

- *Patterns in Mathematics* – number patterns and series
- *Perimeter and Area*
- *Prime Time* – primes, factors, HCF and LCM
- *Fractions*

Class 7 · Ganita Prakash

- *Working with Fractions* – fractions
- *Finding Common Ground* – primes, factors, HCF and LCM
- *Finding the Unknown* – simple equations

Class 7 · Curiosity (science)

- *Measurement of Time and Motion* – speed, distance and time

Class 8 · Ganita Prakash

- *A Square and A Cube* – squares and cubes
- *Proportional Reasoning-1* – ratio and proportion
- *Fractions in Disguise* – percentages, profit and loss, discounts, interest
- *Proportional Reasoning-2* – ratio and proportion
- *Tales by Dots and Lines* – averages (mean)

Reasoning, not a textbook chapter: Clocks and calendars · Number codes and puzzles · Everyday units · Multi-step word problems · Time and work, pipes and tanks · Problems on ages · Mixtures

English

Grammar, vocabulary and reasoning; nothing from the English reader: Vocabulary: meaning in context, opposites, idioms · Grammar: agreement, prepositions, tenses, linking words · Completing and ordering sentences · Analogies, odd one out, letter series and codes · Reading: inference, tone, assumptions, judging an argument · Logical deduction: statements, arrangements, family relations

Science

Class 6 · Curiosity

- *Exploring Magnets* – magnets
- *A Journey through States of Water* – evaporation and condensation

Class 7 · Curiosity

- *Exploring Substances: Acidic, Basic and Neutral* – acids, bases and indicators
- *Electricity: Circuits and their Components* – electric circuits
- *Changes around Us: Physical and Chemical* – physical and chemical changes, rusting
- *Heat Transfer in Nature* – heat and conductors
- *Measurement of Time and Motion* – speed
- *Life Processes in Animals* – respiration
- *Life Processes in Plants* – photosynthesis and plant respiration
- *Light: Shadows and Reflections* – light, shadows and plane mirrors

Class 8 · Curiosity

- *Exploring Forces* – floating, sinking and density
- *Pressure, Winds, Storms, and Cyclones* – force and pressure, pressure in liquids
- *Particulate Nature of Matter* – evaporation and condensation
- *How Nature Works in Harmony* – food chains

Reasoning, not a textbook chapter: Reading an experiment and fair tests · Applying a principle to a new situation

Class 8 syllabus questions

Five maths and five science questions come from these Class 8 chapters only: the first-term chapters, because the test can be taken before the year ends.

Maths

Class 8 · Ganita Prakash

- *Ch 1. A Square and A Cube* – squares, cubes, roots
- *Ch 2. Power Play* – exponents and their laws
- *Ch 3. A Story of Numbers* – number systems
- *Ch 4. Quadrilaterals* – angles and properties of quadrilaterals
- *Ch 5. Number Play* – divisibility
- *Ch 6. We Distribute, Yet Things Multiply* – expanding brackets, identities
- *Ch 7. Proportional Reasoning-1* – ratio, proportion, unit rates

Science

Class 8 · Curiosity

- *Ch 2. The Invisible Living World: Beyond Our Naked Eye* – microorganisms, their uses and harms
- *Ch 3. Health: The Ultimate Treasure* – disease, immunity, vaccines
- *Ch 4. Electricity: Magnetic and Heating Effects* – electromagnets, fuses, heating
- *Ch 5. Exploring Forces* – contact and non-contact forces, friction
- *Ch 6. Pressure, Winds, Storms, and Cyclones* – pressure, air pressure, winds
- *Ch 7. Particulate Nature of Matter* – states of matter, diffusion

Sample paper

These questions were written for this sheet. The real test has different questions, in the same style and at a similar level.

1. MATHS · BASICS
   How many seconds are there in 2.5 hours?

   A 150 **B** 900
   C 9,000 **D** 15,000

2. ENGLISH · BASICS
   Choose the word most nearly opposite in meaning to "reluctant".

   A eager **B** slow
   C afraid **D** unwilling

3. SCIENCE · BASICS
   In the food chain grass → grasshopper → frog → snake, which living thing is the producer?

   A Frog **B** Snake
   C Grasshopper **D** Grass

4. MATHS · CLASS 8
   What is the cube of 5?

   Type-in answer: _____

5. ENGLISH · BASICS
   Choose the correct word: She has been living in Pune ___ 2019.

   A from **B** since
   C for **D** by

6. SCIENCE · CLASS 8
   Some antibiotics, such as penicillin, are made from

   A Fungi **B** Viruses
   C Mosquitoes **D** Algae

7. MATHS · BASICS
   The ratio of boys to girls in a class is 4 : 5. If there are 20 boys, how many girls are there?

   A 16 **B** 20
   C 24 **D** 25

8. ENGLISH · BASICS
   When the coach said the match was "in the bag", he meant that

   A the team had packed its kit **B** the result was still uncertain
   C victory was almost certain **D** the match had been called off

9. SCIENCE · BASICS
   A card with a word printed on it is held in front of a plane mirror. In the mirror, the word appears

   A upside down **B** with left and right swapped
   C larger than on the card **D** smaller than on the card

10. MATHS · CLASS 8
    What is the value of 10^0?

    A 0 **B** 1
    C 10 **D** It has no value

11. ENGLISH · BASICS
The soldier's courage under fire was remarkable. As used here, "remarkable" most nearly means

A ordinary
B doubtful
C hidden
D striking

12. SCIENCE · CLASS 8
Malaria is spread from person to person by

A houseflies
B mosquitoes
C cockroaches
D rats

13. MATHS · BASICS
When a number is doubled and then increased by 7, the result is 31. What is the number?

Type-in answer: _____

14. ENGLISH · BASICS
Choose the correct word: Neither the captain nor the players ___ ready.

A was
B is
C were
D has

15. SCIENCE · BASICS
Compared with the air we breathe in, the air we breathe out contains

A more carbon dioxide
B more oxygen
C less water vapour
D no carbon dioxide at all

16. MATHS · CLASS 8
What do the four angles of any quadrilateral add up to, in degrees?

Type-in answer: _____

17. ENGLISH · BASICS
To "break the ice" at a meeting means to

A make people feel at ease
B damage something by mistake
C cool a drink for guests
D end a long friendship

18. SCIENCE · CLASS 8
A force of 60 N acts on an area of 3 m^2. What is the pressure, in pascals?

Type-in answer: _____

19. MATHS · BASICS
A mother and her daughter are 50 years old together. Five years ago, the mother was 7 times as old as the daughter. How old is the daughter now?

A 8
B 10
C 12
D 15

20. ENGLISH · BASICS
Arrange the sentences in a logical order. P. Asha, their leader, checked the map. Q. The trekkers realised they were lost. R. They reached camp before dark. S. She found the right trail.

A PQSR
B QSPR
C QPRS
D QPSR

21. SCIENCE · BASICS
A sharp knife cuts more easily than a blunt one pushed with the same force, because the sharp edge

A makes much more friction with what is cut
B pushes with a larger force overall
C acts on a smaller area, raising pressure
D spreads the force over a larger area

22. MATHS · CLASS 8

What do you get when you expand $(x + 3)(x + 2)$?

A $x^2 + 5x + 6$ **B** $x^2 + 6$

C $x^2 + 6x + 5$ **D** $2x + 5$

23. ENGLISH · BASICS

What comes next in the series? Z, X, V, T, ___

A S **B** R

C Q **D** P

24. SCIENCE · CLASS 8

Why can a gas be squeezed into a smaller space much more than a solid can?

A Its particles are heavier **B** Its particles do not move

C It has no particles at all **D** Its particles are far apart

25. MATHS · BASICS

A scooter worth ₹80,000 loses 10% of its value every year. What will it be worth after 2 years?

A ₹64,800 **B** ₹64,000

C ₹66,000 **D** ₹72,000

26. ENGLISH · BASICS

Ali, Bina, Dev, Esha and Tara sit in a row of five. Tara is in the middle, Ali at the left end and Bina next to Ali. Dev is not next to Tara. Who is at the right end?

A Dev **B** Esha

C Bina **D** Cannot be determined

27. SCIENCE · BASICS

A box weighing 600 N has one face of area 2 m^2 and another of 0.5 m^2. How many pascals more pressure does it exert resting on the small face than on the large face?

Type-in answer: _____

28. MATHS · CLASS 8

The ratio of boys to girls in a school is 5 : 4. There are 360 students in all. How many are girls?

A 140 **B** 180

C 160 **D** 200

29. ENGLISH · BASICS

All rifles in the store are cleaned weekly. Some weapons in the store are rifles. Which statement must be true?

A All weapons are cleaned weekly **B** Some weapons are cleaned weekly

C No weapon is cleaned weekly **D** Everything cleaned is a rifle

30. SCIENCE · CLASS 8

An electromagnet picks up ten pins. Which change would let it pick up more?

A Use a plastic core **B** Reduce the current

C Add more turns of wire **D** Switch the current off

Answers

1	**C. 9,000**	One hour has 60 × 60 = 3,600 seconds, so 2.5 hours has 2.5 × 3,600 = 9,000 seconds. 150 is the number of minutes.
2	**A. eager**	Reluctant means unwilling or hesitant, so its opposite is eager. "Unwilling" is a word with the same meaning, not the opposite.
3	**D. Grass**	A producer makes its own food by photosynthesis, and in a food chain it always comes first. Grass is the producer; the others are consumers that eat.
4	**125**	The cube of 5 is 5 × 5 × 5 = 125. 25 is its square and 15 is 5 × 3. This is a type-in question.
5	**B. since**	"Since" is used with a point in time (since 2019); "for" is used with a length of time (for five years).
6	**A. Fungi**	Penicillin was first obtained from a mould, a kind of fungus; other antibiotics come from fungi and bacteria. Viruses are not used to make them.
7	**D. 25**	20 boys is 4 parts, so one part is 5 students. Girls are 5 parts, which is 25. 16 comes from using the ratio the wrong way round.
8	**C. victory was almost certain**	"In the bag" is an idiom meaning that success is as good as secured. It has nothing to do with an actual bag.
9	**B. with left and right swapped**	A plane mirror forms an image the same size and the same way up, but with left and right swapped, which is why the word reads backwards.
10	**B. 1**	Any non-zero number raised to the power 0 is 1, because $10^n \div 10^n = 10^0$ and also equals 1.
11	**D. striking**	Remarkable means worth noticing because it is unusual: striking. "Ordinary" is close to its opposite.
12	**B. mosquitoes**	The malaria parasite is carried by the female Anopheles mosquito, which passes it on when it bites. Stagnant water lets mosquitoes breed.
13	**12**	Work backwards: 31 − 7 = 24, and 24 ÷ 2 = 12. Check: 12 × 2 + 7 = 31. This is a type-in question.
14	**C. were**	With "neither … nor", the verb agrees with the nearer subject. "Players" is plural, so the verb is "were".
15	**A. more carbon dioxide**	The body uses oxygen to release energy from food and gives off carbon dioxide, so breathed-out air holds more carbon dioxide and less oxygen.
16	**360**	A diagonal splits a quadrilateral into two triangles, each with angles adding to 180°, so the total is 360°. This is a type-in question.
17	**A. make people feel at ease**	Breaking the ice is getting people talking and relaxed when they first meet. Nothing frozen is involved.
18	**20**	Pressure is force divided by area: 60 N ÷ 3 m^2 = 20 Pa. This is a type-in question: the answer is typed as a number.

19	**B. 10**	Five years ago their ages added to 40, in the ratio 7 : 1, so they were 35 and 5. The daughter is now 10 and the mother 40.
20	**D. QPSR**	The problem comes first (Q), then the response (P), then the solution (S), then the result (R). "She" in S needs Asha named before it.
21	**C. acts on a smaller area, raising pressure**	Pressure is force divided by area. The same force on the thin edge of a sharp knife gives a much greater pressure, which cuts.
22	**A. $x^2 + 5x + 6$**	Multiply each term by each term: $x^2 + 2x + 3x + 6 = x^2 + 5x + 6$. $x^2 + 6$ forgets the two middle terms.
23	**B. R**	Each letter is two places before the one before it: Z, X, V, T, R. S moves back only one place.
24	**D. Its particles are far apart**	Gas particles have large spaces between them, so pushing them closer reduces the volume. In a solid the particles are already packed tightly.
25	**A. ₹64,800**	After one year it is worth ₹72,000. The second year takes 10% of ₹72,000, which is ₹7,200, leaving ₹64,800. ₹64,000 takes ₹8,000 off twice.
26	**A. Dev**	Seats 1, 2 and 3 are Ali, Bina and Tara. Dev and Esha take seats 4 and 5; seat 4 is next to Tara, so Dev must be in seat 5.
27	**900**	On the small face: 600 ÷ 0.5 = 1,200 Pa. On the large face: 600 ÷ 2 = 300 Pa. The difference is 900 Pa. This is a type-in question.
28	**C. 160**	There are 5 + 4 = 9 parts, each of 40 students, so the girls are 4 × 40 = 160. 200 is the number of boys.
29	**B. Some weapons are cleaned weekly**	The weapons that are rifles are cleaned weekly, so at least some weapons are. Nothing is said about the weapons that are not rifles.
30	**C. Add more turns of wire**	More turns of wire, or more current, make an electromagnet stronger. A plastic core in place of iron, or less current, weakens it.